

Slopes of Parallel and Perpendicular Lines

Answer Key

High School Geometry

Quick Answers

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|-------------------------------|--|
| 1. 2 | 6. $\frac{1}{2}, \frac{1}{2}, 2, 2$ |
| 2. -2, $\frac{1}{2}$ | 7. $\frac{1}{2}, -2, 10$ |
| 3. $\frac{3}{2}, \frac{3}{2}$ | 8. $\frac{1}{3}, 3, 4, 4, 4, 4$ |
| 4. $\frac{1}{2}, -2$ | 9. 1, -1, 4.24, 4.24 |
| 5. <i>undefined</i> | 10. 1, 4, 4, 5, $\frac{1}{3}, \frac{1}{3}$ |
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Curriculum

Level: High School Geometry

HSG-GPE.B.4 – *problems 6, 8, 9, 10*

HSG-GPE.B.5 – *problems 1, 2, 3, 4, 5, 7, 10*

HSG-GPE.B.6 – *problems 8, 10*

HSG-GPE.B.7 – *problems 7, 9*

Difficulty 1–5 of 5 across 29 tagged checks.

1. Slope of $\overline{AB}$ for $A(2, 1)$, $B(6, 9)$; then the slope of any parallel line.

Solution: $m_{AB} = \frac{9-1}{6-2} = \frac{8}{4}$.

$\Rightarrow$

$m = 2$

Parallel lines have equal slopes, so every line parallel to $\overline{AB}$ also has slope two.

2. Slope of $\overline{CD}$ for $C(-3, 5)$, $D(1, -3)$; then the slope of any perpendicular line.

Solution: $m_{CD} = \frac{-3-5}{1-(-3)} = \frac{-8}{4}$.

$\Rightarrow$

$m = -2$

A perpendicular slope is the opposite reciprocal: flip the fraction and change the sign, so $-\left(\frac{1}{-2}\right)$ gives the value below. Check: $(-2)\left(\frac{1}{2}\right) = -1 \checkmark$

$\Rightarrow$

$\frac{1}{2}$

3. Is $\overline{PQ}$ with $P(0, -2)$, $Q(4, 4)$ parallel to $\overline{RS}$ with $R(-1, 3)$, $S(3, 9)$?

Solution: $m_{PQ} = \frac{4-(-2)}{4-0} = \frac{6}{4}$.

$\Rightarrow$

$m_{PQ} = \frac{3}{2}$

$$m_{RS} = \frac{9-3}{3-(-1)} = \frac{6}{4}.$$

$\Rightarrow$

$$m_{RS} = \frac{3}{2}$$

The slopes are equal and the segments are not the same line, so $\overline{PQ} \parallel \overline{RS}$.

4. Is $\overline{EF}$ with $E(1, 2)$, $F(5, 4)$ perpendicular to $\overline{GH}$ with $G(0, 6)$, $H(2, 2)$?

Solution: $m_{EF} = \frac{4-2}{5-1} = \frac{2}{4}.$

$\Rightarrow$

$$m_{EF} = \frac{1}{2}$$

$$m_{GH} = \frac{2-6}{2-0} = \frac{-4}{2}.$$

$\Rightarrow$

$$m_{GH} = -2$$

The product of the slopes is $(\frac{1}{2})(-2) = -1$, so the segments are perpendicular.

5. Slope of $\overline{JK}$ for $J(4, -1)$, $K(4, 7)$, and the slope of a perpendicular line.

Solution: $m_{JK} = \frac{7-(-1)}{4-4} = \frac{8}{0}$, and division by zero is not defined, so the segment is vertical and has no slope.

$\Rightarrow$

undefined

Why the opposite-reciprocal rule needs care here (accept any equivalent explanation): the rule $m_1 m_2 = -1$ only applies when both slopes exist as numbers. A vertical segment has no slope to take a reciprocal of. Instead, argue from the geometry: any line perpendicular to a vertical line is horizontal, and a horizontal line has slope zero. So the answer is 0, reached by reasoning about direction rather than by the product rule.

6. $ABCD$ with $A(-2, 1)$, $B(2, 3)$, $C(4, 7)$, $D(0, 5)$: show both pairs of opposite sides are parallel.

Solution: Take the sides in order.

$$m_{AB} = \frac{3-1}{2-(-2)} = \frac{2}{4}, \quad m_{DC} = \frac{7-5}{4-0} = \frac{2}{4}$$

$\Rightarrow$

$$m_{AB} = \frac{1}{2}$$

$\Rightarrow$

$$m_{DC} = \frac{1}{2}$$

$$m_{BC} = \frac{7-3}{4-2} = \frac{4}{2}, \quad m_{AD} = \frac{5-1}{0-(-2)} = \frac{4}{2}$$

$\Rightarrow$

$$m_{BC} = 2$$

$\Rightarrow$

$$m_{AD} = 2$$

$\overline{AB} \parallel \overline{DC}$ and $\overline{BC} \parallel \overline{AD}$, so $ABCD$ is a parallelogram.

7. Triangle PQR with $P(-1, -1)$, $Q(3, 1)$, $R(1, 5)$: show the angle at Q is right, then find the area.

Solution: The angle at Q is formed by $\overline{QP}$ and $\overline{QR}$, so those are the two slopes to compare.

$$m_{PQ} = \frac{1 - (-1)}{3 - (-1)} = \frac{2}{4}$$

$\Rightarrow$

$$m_{PQ} = \frac{1}{2}$$

$$m_{QR} = \frac{5 - 1}{1 - 3} = \frac{4}{-2}$$

$\Rightarrow$

$$m_{QR} = -2$$

The product is -1 , so the two sides meet at a right angle at Q .

Area: with the right angle at Q , the legs are QP and QR , of lengths $\sqrt{20}$ and $\sqrt{20}$, so the area is $\frac{1}{2}\sqrt{20}\sqrt{20} = \frac{1}{2}(20)$ square units.

$\Rightarrow$

10

8. $WXYZ$ with $W(0, 0)$, $X(6, 2)$, $Y(8, 8)$, $Z(2, 6)$: rectangle? And the diagonal midpoints.

Solution (a): Adjacent sides decide squareness of the corners.

$$m_{WX} = \frac{2 - 0}{6 - 0} = \frac{2}{6}, \quad m_{XY} = \frac{8 - 2}{8 - 6} = \frac{6}{2}$$

$\Rightarrow$

$$m_{WX} = \frac{1}{3}$$

$\Rightarrow$

$$m_{XY} = 3$$

Their product is $\left(\frac{1}{3}\right)(3) = 1$, not -1 , so the corner at X is not a right angle and $WXYZ$ is **not** a rectangle. (It is a rhombus: opposite sides are parallel and all four sides have length $\sqrt{40}$.)

Solution (b): Midpoint of diagonal $\overline{WY}$:

$\Rightarrow$

$$(4, 4)$$

Midpoint of diagonal $\overline{XZ}$:

$\Rightarrow$

$$(4, 4)$$

The diagonals share a midpoint, so they bisect each other — which confirms the figure is a parallelogram even though it is not a rectangle.

9. $EFGH$ with $E(1, 2)$, $F(4, 5)$, $G(7, 2)$, $H(4, -1)$: slopes and lengths of $\overline{EF}$ and $\overline{FG}$.

Solution:

$$m_{EF} = \frac{5 - 2}{4 - 1} = \frac{3}{3}$$

$\Rightarrow$

$$m_{EF} = 1$$

$$m_{FG} = \frac{2 - 5}{7 - 4} = \frac{-3}{3}$$

$\Rightarrow$

$$m_{FG} = -1$$

The product is -1 , so the corner at F is a right angle. Now the lengths, by the distance formula:

$$EF = \sqrt{3^2 + 3^2} = \sqrt{18}, \quad FG = \sqrt{3^2 + (-3)^2} = \sqrt{18}$$

$\Rightarrow$

$$EF \approx 4.24$$

$\Rightarrow$

$$FG \approx 4.24$$

What this proves: two adjacent sides are perpendicular and equal in length. Repeating the same two calculations on $\overline{GH}$ and $\overline{HE}$ gives slopes 1 and -1 and the same length, so all four angles are right angles and all four sides are congruent — $EFGH$ is a square.

10. Triangle TUV with $T(0, 0)$, $U(6, 2)$, $V(2, 8)$: the midsegment $\overline{MN}$ and $\overline{TU}$.

Solution (a): Midpoint of $\overline{TV}$, averaging the coordinates:

$\Rightarrow$

$$M(1, 4)$$

Midpoint of $\overline{UV}$:

$\Rightarrow$

$$N(4, 5)$$

Solution (b):

$$m_{MN} = \frac{5 - 4}{4 - 1} = \frac{1}{3}, \quad m_{TU} = \frac{2 - 0}{6 - 0} = \frac{2}{6}$$

Both reduce to the same value.

$\Rightarrow$

$$\frac{1}{3}$$

Solution (c) — model coordinate proof (manual review; accept any complete equivalent): Let M be the midpoint of $\overline{TV}$ and N the midpoint of $\overline{UV}$. By the midpoint formula, M and N have the coordinates found in part (a). Applying the slope formula to M and N gives the slope of $\overline{MN}$, and applying it to T and U gives the slope of $\overline{TU}$; both simplify to the same number. Two distinct lines with equal slopes are parallel, and $\overline{MN}$ and $\overline{TU}$ are distinct because M does not lie on $\overline{TU}$. Therefore $\overline{MN} \parallel \overline{TU}$, which is the triangle midsegment theorem for this triangle.